

Stratum

Owner / operator: Milton Adina Stack: Vercel + Supabase only Prepared: 2026-05-30

Prices verified: vercel.com/pricing · supabase.com/pricing (2026-05-30)

TL;DR

The platform is **built, live, and hardened**. To finish and run it commercially you need exactly two paid plans — **Vercel Pro and Supabase Pro, a flat ~\$45/month** — plus a free Stripe account and a **one-time ~\$20–50** of LLM credits to switch the savings engine on. Your ongoing per-token cost can be **\$0**: each customer brings (or is billed for) their own model usage.

Bottom line: ~\$45 / month recurring + ~\$50 once. That is the entire cost of running this masterpiece.

1 · Where the platform stands today

A commercial, multi-tenant LLM **gateway + optimization** platform, already deployed to your Vercel project and backed by your Supabase database:

- **Multi-tenant API-key auth**, per-plan rate limits, token budgets, concurrent-session caps.
- **Live on Vercel** (`/health` 200, reaching real Supabase).
- **Multi-provider routing** — one Anthropic-compatible endpoint that runs on Anthropic, OpenAI, OpenRouter, Gemini, or a local model, billing the exact upstream token counts.
- **Billing pipeline end-to-end**: usage → signed records → invoice engine → Stripe send → paid-webhook → read API + CFO dashboard.
- **Security**: RLS on all 17 tables, append-only financial ledger, service-role-only access, PII redaction, and **8 adversarial reviews + a full audit / bughunt / pentest** (≈75 confirmed defects fixed, incl. CRITICAL auth-bypass, key-leak, double-charge). Live pentest: the public anon key reads **zero rows**.
- **Test suite: 825 passing**; typecheck + lint green.

It does not yet *prune* tokens in production (the "20% of savings" value-prop) — that is intentionally gated behind a safety eval (\$2B) and needs the two paid plans to be commercial-legal and always-on.

2 · What remains

A · Go-commercial activation — needs the two paid plans + a free Stripe account

STEP	WHAT IT IS	NEEDS
Upgrade Vercel → Pro	Hobby forbids commercial use and caps function duration; Pro unlocks commercial use, long streaming functions, the encrypted env store, and production.	required Vercel Pro · \$20/mo
Upgrade Supabase → Pro	The Free project pauses after 7 days of inactivity (DB goes offline) and has no backups. Pro never pauses; adds 7-day daily backups, 8 GB, more connections.	required Supabase Pro · \$25/mo
Stripe account	Free account → <code>sk_test_</code> → <code>npm run verify-stripe</code> (\$1 round-trip) → flip to <code>sk_live_</code> + register <code>.../stripe/webhook</code> (<code>whsec_</code>).	free 2.9% + 30¢ / charge
First paying customer	A design partner who routes traffic through the proxy and pays the first invoice.	a customer, not a cost

B · Turn on the savings engine ("20% of savings") — needs one-time LLM credits

Pruning is the business. It is held behind a **safety gate** (pruning may not enter the request path until an eval shows **<5% answer-quality degradation**). Clearing it is a one-time eval run with a real judge model; the encoder runs locally for free.

- **Needs**: one-time ~\$20–50 of Anthropic (or other judge-model) API credits.
- **Then**: enable pruning → every customer's bill drops → you invoice 20% of the measured drop.

- **No new infrastructure** — runs on the same Vercel + Supabase.

C · Optional enhancements — no new infrastructure

Already built on Supabase, needing only wiring: the **three-tier memory** (Supabase `pgvector` — **no Pinecone / Neo4j**) and the **git-attestation audit engine** (deterministic tier free; LLM escalation credit-gated). Defer until a customer asks.

D · The one thing Vercel can't host — deferred by design

The **TEE / ZK-Context encryption** feature (encrypt customer context inside a hardware enclave) requires **AWS Nitro Enclaves** — hardware that does not exist on Vercel. It is an enterprise-only add-on and is **not needed for the MVP**: your data is already protected by RLS, service-role-only access, PII redaction, and never storing raw context. Revisit only for an enterprise contract that requires hardware-attested encryption, on a dedicated TEE host at that time.

3 · What you need from each provider

Vercel		\$20/mo	Supabase		\$25/mo
Plan	Pro (per member — you're the only one)		Plan	Pro (includes \$10/mo compute credit covering the Micro instance)	
Why	Commercial use is not allowed on Hobby ; Pro also unlocks the long streaming-function durations the proxy needs + the prod env store you already use.		Why	Free pauses after a week + no backups. Pro = never pauses, 8 GB DB, 7-day daily backups, more connections.	
Do	Team settings → upgrade to Pro. Project's already deployed; nothing else changes.		Do	Org settings → upgrade to Pro. Schema/migrations already applied & verified.	
Usage	Active CPU \$0.128/hr · invocations \$0.60/1M · 1 TB transfer included then \$0.15/GB → ~\$0–10/mo at pilot.		Skip	PITR (+\$100/mo) — the 7-day backups suffice for the MVP. Add later when revenue justifies.	

4 · Cost summary

Recurring (monthly)

ITEM	PROVIDER	BASE	PILOT USAGE	NOTES
Pro plan	Vercel	\$20.00	\$0–10	required for commercial use + long streams
Pro plan	Supabase	\$25.00	~\$0	never-pause + 7-day backups; \$10 credit incl.
Payments	Stripe	\$0.00	per-charge	2.9% + 30¢ per successful charge only
LLM tokens (production)	upstream API	\$0.00	—	customer brings/pays their own key (pass-through)
Recurring total		\$45.00	≈ \$45–65	flat at the start

One-time · and optional (don't buy now)

ITEM	PROVIDER	COST	WHEN
Safety eval to enable pruning	Anthropic API	~\$20–50 once	the only LLM spend you fund yourself
Point-in-Time Recovery <code>optional</code>	Supabase	+\$100/mo	when revenue justifies tighter recovery
TEE encryption host <code>optional</code>	AWS Nitro (not Vercel)	varies	only for an enterprise hardware-attestation contract

5 · How we keep the spend lowest (without cutting corners)

- **Customers fund their own model usage.** The proxy forwards on the customer's own LLM key (or bills it back), so your ongoing per-token cost is **\$0** — you invoice only 20% of the savings you create. The single biggest lever, already designed in.
- **One judge run, not a subscription.** The pruning eval is a one-time spend; the encoder is local & free.
- **Stay inside the included allowances.** Pilot traffic fits Vercel's 1 TB transfer and Supabase's 250 GB egress / 8 GB DB — so the only fixed cost is the two base plans.
- **No fourth vendor.** Memory uses Supabase `pgvector` (not Pinecone), the graph uses Supabase tables (not Neo4j); no separate queue/cache.

- **Skip PITR and TEE until they pay for themselves.** Both are real but neither is needed to ship or to be secure.

6 · Activation checklist (in order)

- ☐ **Upgrade Vercel** → **Pro** (Team → Plans).
- ☐ **Upgrade Supabase** → **Pro** (Organization → Billing).
- ☐ Create a **Stripe** account → copy the `sk_test_` key.
- ☐ Run `npm run verify-stripe` (\$1 test invoice end-to-end). On pass → switch to `sk_live_` and register the `/stripe/webhook` endpoint to get `whsec_`.
- ☐ Add `STRIPE_SECRET_KEY`, `STRIPE_WEBHOOK_SECRET` (and, if self-funding, an upstream LLM key) to **Vercel's env store** → redeploy.
- ☐ **Onboard the first partner:** `npm run create-org` → give them their CQ key → point their tool's `ANTHROPIC_BASE_URL` at your proxy.
- ☐ **When monetizing savings:** fund the one-time eval, run it green, enable pruning, then `npm run invoice -- --org-id <them> --send`.

Reflects the platform at commit `0dde17e` (branch `stratum-phase-0-capture`): v1.0.0 commercial foundation green except the two operator-provided items (a Stripe key and a paying partner), multi-provider gateway shipped and adversarially reviewed, 825 tests passing, deployed and verified live. · Stratum — token-arbitrage LLM gateway.